

PRODUCT REQUIREMENTS DOCUMENT

Medical Appointment No-Show Prediction System

Document Type	Product Requirements Document (PRD)
Product Name	Medical Appointment No-Show Prediction System
Version	1.0 (Draft)
Status	Proposed
Owner	Product / Data Science Team

1. Overview

Missed medical appointments — patient **no-shows** — are a persistent operational problem for hospitals, clinics, and diagnostic centers. Every unattended slot represents wasted clinician time, longer waitlists for other patients, and direct revenue loss.

This PRD defines the requirements for a **Medical Appointment No-Show Prediction System**: a machine learning-powered product that scores each upcoming appointment with a probability of no-show, so that clinic staff can intervene early (reminders, calls, rescheduling, overbooking) for the appointments most at risk.

2. Problem Statement

Clinic staff currently have no reliable, data-driven way to know in advance which patients are unlikely to show up. Reminder calls, SMS, and follow-ups are applied uniformly (or not at all), wasting staff effort on low-risk patients while high-risk no-shows go unaddressed. Historical appointment data is collected but not used to predict future attendance behavior.

3. Goals and Success Metrics

Business goals:

- Reduce the overall appointment no-show rate.
- Improve utilization of clinician time and appointment slots.
- Enable targeted, cost-effective reminder and outreach strategies.
- Give administrators visibility into no-show patterns through dashboards.

Success metrics (KPIs):

Metric	Definition / Target
No-show rate reduction	Decrease in overall no-show % after rollout, measured monthly
Model recall	Proportion of actual no-shows correctly flagged (target: high recall, since missed high-risk cases are costly)
Model precision / F1-score	Balance of correct high-risk flags vs. false alarms
Adoption rate	% of clinics/staff actively using the risk scores in daily workflow
Intervention lift	No-show rate for flagged patients who received an intervention vs. those who did not

4. Target Users

User	Needs / Use of the System
Clinic Administrator	Views dashboards, monitors no-show trends, plans staffing and overbooking

User	Needs / Use of the System
Front-desk / Scheduling Staff	Sees per-appointment risk scores, prioritizes reminder calls/SMS
Doctor / Care Provider	Optionally sees aggregate no-show risk for their daily schedule
Data/ML Team	Trains, monitors, and retrains the prediction model

5. Scope

In scope:

- Ingestion and preprocessing of historical appointment data.
- A machine learning model predicting Show / No-Show probability per appointment.
- A scoring pipeline that runs on new/upcoming appointments.
- Basic dashboards/visualizations of no-show patterns (age, gender, weekday, SMS effect, etc.).
- Model evaluation and comparison across multiple algorithms.

Out of scope (for v1):

- Automated sending of SMS/calls (v1 only surfaces risk scores; outreach remains manual).
- Real-time integration with third-party hospital scheduling systems.
- Mobile application (web dashboard only in v1).
- Deep-learning models (reserved for future iterations).

6. Functional Requirements

ID	Requirement
FR-1	The system shall ingest historical appointment records including patient demographics, scheduled/appointment dates, SMS reminder status, and prior attendance history.
FR-2	The system shall clean and preprocess data: remove duplicates, handle missing values, encode categorical fields, and derive date-based features.
FR-3	The system shall engineer features such as waiting days, day of week, appointment month, age group, and a patient's historical no-show rate.
FR-4	The system shall train and compare multiple classification models (e.g., Logistic Regression, Decision Tree, Random Forest, XGBoost, SVM) to predict no-show probability.
FR-5	The system shall output a no-show probability score (e.g., "78% probability of No-Show") for each upcoming appointment.
FR-6	The system shall allow staff to view a ranked list of upcoming appointments by no-show risk.
FR-7	The system shall provide visual analytics (charts/dashboards) showing no-show rate by age, gender, weekday, SMS reminder status, and waiting time.

ID	Requirement
FR-8	The system shall support periodic retraining of the model as new appointment data becomes available.
FR-9	The system shall log model predictions and actual outcomes to support ongoing evaluation.

7. Non-Functional Requirements

Category	Requirement
Performance	Scoring an upcoming appointment should return a prediction in near real time (seconds, not minutes).
Scalability	The pipeline should handle growing historical datasets (tens of thousands of records or more) without redesign.
Accuracy	The chosen model should be evaluated on Accuracy, Precision, Recall, F1-score, Confusion Matrix, and ROC-AUC, with Recall/F1 prioritized.
Data Privacy & Security	Patient health data must be handled per applicable privacy regulations, with access controls and secure storage.
Usability	Dashboards must be understandable by non-technical clinic staff and administrators.
Maintainability	The model and pipeline should be modular enough to retrain and redeploy with minimal manual effort.

8. Data Requirements

Core dataset fields required to build and run the model:

Field	Description
Patient ID	Unique patient identifier
Appointment ID	Unique appointment identifier
Gender	Patient's gender
Age	Patient's age
Scheduled Date	Date when appointment was scheduled
Appointment Date	Actual appointment date
Waiting Days	Days between scheduling and appointment
SMS Received	Whether reminder SMS was received
Previous No-Shows	Number of previous missed appointments
Previous Appointments	Number of previous appointments
Chronic Condition	Whether patient has a chronic condition
Appointment Type	Type of medical appointment

Field	Description
Day of Week	Day of scheduled appointment
No-Show (target)	Whether the patient missed the appointment

9. Technical Approach

Component	Choice
Language	Python
Core Libraries	Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn
Database	MySQL / SQL
Visualization	Power BI / Tableau / Python (Matplotlib, Seaborn)
Dev Environment	Jupyter Notebook / Google Colab / VS Code

Model development will follow: data collection → preprocessing → exploratory data analysis → feature engineering → model training and comparison → evaluation → deployment of the scoring pipeline.

10. Assumptions and Constraints

- Historical appointment data of sufficient volume and quality is available for training.
- Reasonable proxies exist for factors like SMS reminders and prior attendance history.
- The model produces a probability, not a certainty — final intervention decisions remain with staff.
- Patient behavior may shift over time, requiring periodic retraining.

11. Risks and Mitigations

Risk	Mitigation
Poor data quality (missing/inconsistent fields)	Invest in a robust preprocessing pipeline; validate data at ingestion.
Model bias against certain patient groups	Monitor fairness metrics across demographic slices; audit regularly.
Low staff adoption of risk scores	Keep the interface simple; integrate scores into existing scheduling workflow.
Model drift over time	Schedule periodic retraining and performance monitoring.
Privacy/compliance concerns with health data	Apply access controls, encryption, and regulatory review before rollout.

12. Future Scope (Roadmap)

- Integrate with real-time hospital appointment/scheduling systems.
- Automate reminder outreach (SMS/calls) for high-risk patients.
- Build a dedicated web or mobile application for staff.
- Explore deep-learning models for improved accuracy.
- Expose a real-time prediction API for third-party integration.
- Embed predictions directly into Power BI dashboards.
- Set up continuous model retraining as new data arrives.

13. Conclusion

This PRD defines a focused, data-driven product that turns historical appointment records into actionable no-show risk scores. By prioritizing high-risk appointments for intervention, the system aims to reduce missed appointments, improve resource utilization, and give healthcare providers a repeatable, measurable process for managing appointment attendance.